BRANDGRAPH AI

An Evidence-Linked Brand Intelligence and Governance System

Inkloom AI Brand Governance Platform Challenge

Round 1

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Concept Comparison: BrandShield AI vs BrandGraph AI

Factor	**BrandShield AI**	**BrandGraph AI**
Problem coverage	General brand review and correction	Brand knowledge + evidence-grounded review
Technical depth	Rule checks + optional LLM	Rule checks + RAG + graph relationships + evaluation
Explainability	Rule-linked findings	Source-linked findings with graph context
Risk	Lower integration risk	Higher risk, mitigated by local graph and optional API
MVP feasibility	High	High if text-first and graph database optional
Best use	Safe fallback	Recommended competition concept

So we selected BrandGraph AI.

BrandGraph AI with a text-first MVP and a built-in Brand Stress Tester. This raises technical depth while keeping the first release feasible. BrandShield AI remains the lower-risk fallback.

1. Executive Summary

Today, organisations share content across websites, social media, marketing campaigns, presentations, product pages and partner platforms. With more content creators, agencies and AI tools involved, maintaining a consistent brand identity has become increasingly difficult. Content may contain outdated product information, unapproved terms, unsupported claims, missing disclaimers or a tone that does not match the brand's identity.

BrandGraph AI is a proposed AI-powered platform designed to help organisations identify and correct these issues before content is published. It brings approved brand guidelines, product information and supporting evidence together in a connected knowledge graph. When a user submits a draft, the system checks it against the available rules, retrieves relevant evidence and uses AI-assisted analysis where contextual judgement is needed, such as evaluating tone and messaging consistency.

One of the main goals of BrandGraph AI is to make every finding understandable and traceable. Instead of simply flagging an issue, the platform explains what is wrong, identifies the relevant rule or source, and suggests how the content could be improved. It also distinguishes confirmed violations from possible inconsistencies and claims that need further verification. Final decisions remain with authorised human reviewers.

The initial version is planned as a text-based prototype with a local knowledge graph, searchable brand information, rule-based checks, an evidence-linked review interface, a basic approval workflow and a Brand Stress Tester. More advanced features, such as a graph database and visual-content analysis, can be explored in future versions. These features are proposed development goals and are not presented as already completed.

2. Problem Definition

Organisations often struggle to maintain a consistent brand identity across different platforms and communication channels. As highlighted in the challenge brief, common issues include inconsistent messaging, incorrect or outdated product information, inappropriate terminology, off-brand visuals and promotional claims that lack supporting evidence.

Another challenge is that brand guidelines and approved information are often spread across multiple documents and shared folders. Content creators may find it difficult to identify the correct rule or determine whether a particular claim is still valid. As a result, reviewers spend additional time checking content manually, and different reviewers may provide different feedback on similar issues.

The main problem addressed by BrandGraph AI is:

How can content teams identify and resolve brand-governance issues before publication while ensuring that every finding is supported by evidence, clearly explained and reviewed by a human when necessary?

The problem definition is based on the challenge brief. Interviews, surveys, user observations and usability tests have not been claimed as completed. If time permits, the team should validate these challenges by speaking with content creators, marketing professionals or brand reviewers and documenting the actual findings.

3. Target Users and Their Needs

BrandGraph AI is intended for people who create, manage or review brand-related content.

The primary target users include:

- **Marketing and content teams:** To ensure that campaigns and other communications follow approved brand guidelines.
- **Brand managers:** To maintain consistency in terminology, tone, messaging and product information.
- **Agencies and freelance creators:** To check whether their content meets a client's requirements before submitting it for approval.
- **Communications and compliance reviewers:** To identify unsupported claims, missing disclaimers and violations of approved policies.

The prototype aims to address several practical needs. Users should be able to find relevant guidelines quickly, identify questionable statements, understand why an issue was flagged, and improve their drafts without losing the original message. They should also be able to track revisions and review decisions, while brand managers can examine recurring issues to identify areas where guidelines may need clarification.

The first version will focus on a single fictional brand and text-based content. Features such as managing multiple brands, enterprise-level identity management and direct publishing integrations are outside the initial scope.

4. Proposed Solution and Workflow

BrandGraph AI follows a structured workflow that connects brand knowledge, automated checks and human review.

Step 1: Set up brand knowledge

The user creates a sample brand profile by adding approved guidelines, preferred and prohibited terms, product facts, supporting sources and mandatory disclaimers. The system organises this information into a structured knowledge model and makes the source documents searchable.

Step 2: Submit content for review

A content creator submits a draft, such as a marketing post, product description or campaign message. The system analyses the text against the available brand requirements.

Step 3: Check rules and retrieve evidence

The rule engine checks explicit requirements, including prohibited phrases and mandatory disclaimers. At the same time, the retrieval component searches approved sources for relevant facts and guidelines. The knowledge graph helps connect products, claims, policies, terminology and evidence sources.

Step 4: Analyse and explain potential issues

Where contextual analysis is needed, an optional AI component evaluates aspects such as tone and messaging consistency. The system then presents findings with the relevant text, supporting rule or evidence, an explanation and a suggested action.

Each finding is classified appropriately as a confirmed rule violation, a potential inconsistency or a claim that needs verification.

Step 5: Suggest corrections and collect feedback

The platform suggests alternative wording that follows the brand guidelines while preserving the original meaning. The creator can accept, edit or reject the suggestion. An authorised reviewer can add comments, request changes, approve the draft or reject it.

Step 6: Track revisions and evaluate performance

A basic revision history records drafts, findings and review statuses. The Brand Stress Tester runs synthetic test cases to evaluate how the system handles clear violations, compliant content, conflicting rules, missing evidence and ambiguous claims.

Content is never published automatically as part of this workflow. If a claim lacks supporting evidence, the system does not automatically assume that the claim is false. Instead, it marks the claim as **Needs verification**, unless approved evidence establishes a contradiction.

5. Core Features and Differentiation

A. Brand Knowledge Graph

The knowledge graph organises brand information into connected entities, including Brand, Product, Rule, Term, Claim, EvidenceSource, Disclaimer and ContentDraft.

Relationships such as SUPPORTS, PROHIBITS, REQUIRES, APPLIES_TO and SUPERSEDES help the system understand how different pieces of information relate to one another. Source references are retained wherever applicable so that findings can be traced back to their origin.

B. Retrieval-Augmented Generation (RAG)

The RAG component retrieves relevant passages from approved brand documents and combines them with connected knowledge-graph information. This helps ground the review in available evidence rather than relying entirely on the AI model's general knowledge.

When the available sources do not provide enough information, the system should acknowledge the limitation instead of inventing brand rules or product facts.

C. Deterministic Rule Engine

The rule engine checks requirements that can be expressed explicitly, such as preferred terminology, prohibited phrases, mandatory disclaimers and predefined claim restrictions.

These checks produce predictable results and do not depend on an LLM. This makes them particularly useful for requirements where consistency and reliability are important.

D. Explainable Findings

Each finding includes the relevant text excerpt, issue category, severity where appropriate, supporting rule or source, a brief explanation and a recommended next step.

The platform distinguishes between three main outcomes:

- **Confirmed rule violation:** The content directly conflicts with an explicit, applicable rule.
- **Potential inconsistency:** The content may conflict with the brand's intended tone or messaging and requires further judgement.
- **Needs verification:** The available evidence is insufficient to confirm whether the claim is correct.

This distinction helps reviewers understand the difference between a definite violation and an issue that requires additional investigation.

E. Correction Assistant

The correction assistant proposes alternative wording for flagged content. Its objective is to improve brand compliance while preserving the creator's original intent.

Users remain in control of the final wording and can accept, modify or reject each suggestion.

F. Human Review Workflow

The platform supports a basic review process with the following statuses: Draft, Needs Revision, Submitted for Review, Approved and Rejected.

Reviewers can provide comments, request modifications and record decisions. A basic revision history helps the team understand how content changes during the approval process.

G. Brand Stress Tester

The Brand Stress Tester evaluates the system using synthetic examples designed to represent different review situations. These include clear policy violations, compliant drafts, conflicting guidelines, missing evidence and ambiguous claims.

The tester records the system's observed responses so that the team can identify missed violations, incorrect flags and cases where the system should have expressed greater uncertainty. Performance claims should be made only after the tests have actually been executed and their results recorded.

H. Recurring-Issue Insights

The platform can summarise stored findings by category, rule or issue type. These summaries may help brand managers identify frequently encountered problems and determine whether certain guidelines need clarification.

Any statistics or performance metrics must be calculated from actual recorded data. The prototype should not present estimated or illustrative values as measured results.

Overall Differentiation

BrandGraph AI aims to combine a connected brand knowledge graph, evidence-based retrieval, deterministic policy checks, transparent explanations and human-controlled review in a single workflow.

Its key design principle is **traceability with human oversight**. Rather than simply labelling content as compliant or non-compliant, the platform is intended to show why an issue was identified, what evidence supports the finding and what action the user can take next.

These are the proposed differentiators of the system. Their effectiveness and advantages over existing approaches must be evaluated through implementation and testing rather than assumed in advance.

6. Technical Architecture

BrandGraph AI will use a Python-based architecture designed to keep the initial prototype simple, modular and easy to test. Streamlit will serve as the frontend, providing an interactive dashboard where users can submit content, review findings, explore evidence and manage approvals.

The application will be divided into separate modules so that each component has a clear responsibility.

Module	Responsibility
app.py	Main Streamlit interface and navigation
ingestion.py	Import and process brand guidelines and source documents
graph_store.py	Create, query and manage knowledge-graph relationships
retrieval.py	Search approved documents and retrieve relevant evidence
rule_engine.py	Apply deterministic brand-policy checks
ai_review.py	Perform optional AI-assisted contextual analysis
stress_tests.py	Run synthetic test cases and record outcomes
workflow.py	Manage review states, comments and decisions
storage.py	Handle persistent storage and database operations
tests/	Contain unit, integration and end-to-end tests

Knowledge graph: The first version will use a lightweight local graph built with NetworkX or a validated node-and-edge schema stored in JSON or SQLite. A common GraphStore interface will keep the design flexible enough to support Neo4j later if the project requires it. A separate graph database will not be necessary for the initial demonstration.

Search and retrieval: Brand documents will be divided into smaller text sections, with each section linked to a source ID and, where available, a section title. Keyword-based retrieval will be implemented first because it is straightforward to test and does not require an external service. Semantic embeddings may be introduced later if the available time and deployment environment permit. Retrieved passages will be displayed with their source labels.

AI analysis: An optional language-model API may be used to evaluate contextual issues, such as tone, messaging consistency and alternative wording. Responses will follow a defined output schema and undergo validation before being displayed. The implementation will include input limits, timeouts, error handling and a fallback path. AI-generated findings will be presented as suggestions rather than verified facts.

Data storage: SQLite will store content drafts, rules, findings, comments, review statuses and revision records. API credentials will be supplied through environment variables or an appropriate secrets manager rather than embedded in source code.

Testing: The project will include unit tests for rule checks and graph relationships, integration tests for retrieval and workflow operations, and synthetic end-to-end scenarios. Negative tests will cover missing evidence, contradictory rules, empty inputs, malformed model responses and unavailable APIs.

7. Proposed Data Model

The knowledge graph will organise brand information into defined node types. Each node represents an entity or record that the system needs to understand, while edges describe the relationships between them.

The proposed node types are Brand, Product, Rule, PreferredTerm, ProhibitedTerm, Claim, EvidenceSource, Disclaimer, ContentDraft, Finding and ReviewDecision.

Examples of relationships include:

- Product — HAS_APPROVED_NAME → PreferredTerm
- EvidenceSource — SUPPORTS → Claim
- Rule — REQUIRES → Disclaimer

- Rule — SUPERSEDES → Rule
- Finding — TRIGGERED_BY → Rule
- Finding — SUPPORTED_BY → EvidenceSource

These relationships will help the platform connect a content issue to the relevant brand requirement and supporting information.

Each source-backed record should contain a stable identifier, title, relevant excerpt and provenance details. An effective date should also be included when it is available. If the source or date is unknown, the system will represent that uncertainty instead of filling the gap with an assumption.

When two rules conflict or an older rule may have been replaced by a newer one, the system should flag the situation for human clarification rather than silently selecting one rule.

8. Responsible Design, Privacy and Security

BrandGraph AI will be designed to handle brand information carefully and communicate its limitations clearly.

The public demonstration will use fictional brand data unless permission has been obtained to use real organisational material. Personal information and confidential content will be minimised, and users will be informed when submitted content is sent to an external AI service. Sensitive material will not be sent without appropriate authorisation.

API keys and other credentials will be kept out of source control. The repository will include suitable .gitignore rules, while uploaded files and user inputs will be validated. File types and sizes will be restricted to the extent implemented, and access to drafts will be controlled according to the application's actual capabilities.

The system will provide source-linked explanations wherever supporting evidence is available. It will also communicate uncertainty, allow users to correct or dismiss findings, and retain human approval as part of the review process.

Any security measures described in the final submission must match the safeguards that have actually been implemented. The project will not claim enterprise-grade security, formal certification or compliance with a particular standard without supporting evidence.

The team will also disclose significant AI-generated components, third-party models, libraries, APIs, templates and external services in accordance with the challenge requirements and applicable licences.

9. Development Plan and Scope Control

Development will follow a series of milestones, with each stage building on a working baseline.

Milestone 1 — Establish the baseline

Create the Streamlit application, add a fictional brand dataset, define the initial rule schema and build the main navigation.

Milestone 2 — Build the knowledge graph

Implement nodes, edges and source provenance. Add relationship visualisation and basic handling of conflicting or superseded rules.

Milestone 3 — Implement rules and retrieval

Develop explicit policy checks, keyword-based retrieval and source-linked findings. Test the workflow using synthetic examples.

Milestone 4 — Add optional AI analysis

Integrate structured language-model analysis only after the core workflow is functional. Validate model outputs and provide a fallback for cases where no API key is configured or the service is unavailable.

Milestone 5 — Complete the review workflow and stress tests

Add review states, comments and revision records. Create the synthetic test suite and generate a summary of the actual test results.

Milestone 6 — Test and prepare the submission

Run the automated tests, fix identified defects, document the architecture, prepare the README, disclose external components and assemble a demonstration using working features.

The main priority is to deliver a reliable end-to-end text-review workflow. If development time becomes limited, graph relationships, evidence-linked rules and stress testing will take precedence over visual analysis, complex agent orchestration and production integrations.

10. Evaluation Plan and Acceptance Criteria

The prototype will be evaluated using a labelled synthetic test set. Each test case will define the input content, expected issue category, relevant rule or source, and expected outcome.

The planned test scenarios include:

1. Content containing a prohibited phrase.
2. Use of an unapproved product name.
3. A missing mandatory disclaimer.
4. Content that follows all applicable rules.
5. A claim supported by approved evidence.
6. A claim contradicted by approved evidence.
7. A claim for which no supporting evidence is available.
8. Two conflicting brand rules.
9. An empty content submission.
10. A malformed response from the language model.
11. An unavailable AI service or missing API configuration.

The proposed acceptance criteria are as follows:

- Explicit violations are detected in the curated test cases.
- Every evidence-backed finding references an existing rule or source identifier.
- Claims without sufficient evidence are labelled “Needs verification.”
- Users can edit a draft and complete the relevant review-state transitions.
- The core application works without an AI API key.
- Automated tests execute successfully, with their outcomes documented.

The team will record actual test counts, false positives, missed violations and failures after running the test suite. Accuracy percentages will be reported only when they have been calculated from a defined test set and actual results.

Synthetic testing can demonstrate that the implementation behaves as expected in selected scenarios, but it cannot establish real-world effectiveness on its own. Further validation with actual content reviewers would be needed to assess usability and practical value.

11. Risks and Mitigation

Several technical and operational risks may affect the development and reliability of BrandGraph AI.

Risk	Proposed mitigation
AI generates unsupported findings	Ground explanations in retrieved evidence, validate outputs and display uncertainty.
Knowledge-graph implementation becomes too complex	Start with a small schema and local graph before considering a separate graph database.
Retrieval returns irrelevant or incomplete evidence	Preserve source information, display retrieved excerpts and test cases with missing evidence.
AI API becomes unavailable or expensive	Keep deterministic checks and local keyword search functional without the API.
Additional features delay the core product	Prioritise text review and defer visual analysis and external integrations.
Brand guidelines conflict or remain ambiguous	Flag the conflict for authorised human clarification rather than automatically choosing a rule.

These mitigations are intended to reduce implementation risk while keeping the prototype manageable. Their effectiveness will depend on how successfully they are implemented and tested.

12. Expected Impact and Future Roadmap

If implemented and validated successfully, BrandGraph AI could help content creators find relevant brand requirements more quickly, enable reviewers to explain decisions consistently and help brand managers identify recurring violations or unclear guidelines.

The expected benefits are based on the proposed functionality and have not yet been measured. Any claims about time savings, improved consistency or reduced review effort will require evidence from actual testing.

Future development could explore additional capabilities, including:

- Document-level brand compliance reviews.
- Image and logo checks, including colour consistency.
- Accessibility checks for digital content.
- Integration with content management and publishing systems.
- More advanced access controls and multi-brand support.
- Expanded analytics for recurring governance issues.

These capabilities are possible future extensions rather than commitments for the initial prototype.

13. Conclusion

BrandGraph AI proposes a structured approach to brand governance by combining connected brand knowledge, evidence-based retrieval, deterministic rule checks and optional AI-assisted analysis.

The knowledge graph connects approved brand information, while retrieval identifies the evidence relevant to a submitted draft. The rule engine detects explicit violations, and the optional AI component helps assess contextual issues and suggest improvements. A review workflow and synthetic stress tests provide additional ways to examine the system's behaviour.

A central principle of the project is that every important finding should be understandable and traceable, with human reviewers retaining final authority over approval decisions.

The text-first approach keeps the initial implementation achievable while still demonstrating meaningful technical depth. The project's success should be assessed through its working end-to-end workflow, evidence traceability, handling of uncertainty, actual test results, usability and quality of documentation—not through unverified claims of accuracy or completion.

14. Research, Implementation and Disclosure Statement

This document describes the proposed design, implementation plan and evaluation approach for BrandGraph AI. It does not claim that user interviews, surveys, implementation milestones, testing activities, productivity improvements or deployment have already been completed unless these can be independently verified.

Before submission, the team should confirm the challenge portal's page limit, required template, deadline and attachment requirements. Repository history and team contributions should be documented accurately. Significant AI assistance and relevant third-party components should be disclosed, and demonstration data should be fictional or authorised for use.

Appendix A. Suggested Demonstration Script

The demonstration should follow the complete workflow, from configuring brand knowledge to reviewing a draft and evaluating the system.

1. **Open the brand workspace.** Show the fictional brand, approved rules, products, claims and supporting evidence sources.
2. **Submit a test draft.** Include a terminology violation, a missing disclaimer and an unsupported claim.
3. **Review the findings.** Demonstrate the rule-based checks and the evidence retrieved for each issue.
4. **Explain uncertainty handling.** Compare a confirmed violation with a claim that requires verification.
5. **Complete a review cycle.** Edit the draft, submit it for review, add a reviewer comment and record a decision.
6. **Run the Stress Tester.** Display the actual pass/fail outcomes produced by the synthetic test suite.
7. **Discuss limitations.** Explain external AI usage, known limitations and features that remain on the roadmap.

Only features that work in the submitted version should be demonstrated as implemented.

Appendix B. Synthetic Demonstration Data

The demonstration will use a clearly identified fictional company, fictional products and synthetic brand guidelines. The sample dataset can include approved terminology, prohibited terms, mandatory disclaimers, fictional product facts and supporting evidence snippets.

Invented sources should be labelled as “Synthetic demo source” so that they cannot be mistaken for real-world evidence.

For example, a fictional guideline might state: “Use Product Nova as the approved product name; do not use Nova Pro unless authorised.”

A test draft could contain the claim: “Product Nova reduces processing time by 30%.” This statement must be explicitly labelled as a fictional test fixture rather than a verified product fact. The system should then demonstrate how it handles the claim according to the evidence available in the synthetic dataset.

Selected Technical References

- **Streamlit documentation:** https://docs.streamlit.io/
- **Neo4j GraphRAG for Python documentation:** https://neo4j.com/docs/neo4j-graphrag-python/current/

These references provide guidance for the proposed technology choices. Their inclusion does not imply that Streamlit extensions, Neo4j or Neo4j GraphRAG have already been integrated into the prototype.